

L03_Elizabeth_Carrillo_ITAI3377

February 13, 2026

```
[1]: import tensorflow as tf
from tensorflow.keras import layers, models
import os

# --- L03: Elizabeth Carrillo Clavijo ---
# This code matches the reflection submitted on Feb 7 [cite: 2, 3]

# 1. Dataset and Preprocessing
# Normalizing and reshaping to (28, 28, 1) as documented [cite: 9]
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data()
x_train = x_train.reshape((-1, 28, 28, 1)).astype('float32') / 255.0
x_test = x_test.reshape((-1, 28, 28, 1)).astype('float32') / 255.0

# 2. Model Training
# Architecture: Conv2D (32 filters) -> MaxPooling -> Dense (128) -> Softmax
# ↳ (10) [cite: 11, 12, 15, 16]
model = models.Sequential([
    layers.Conv2D(32, (3, 3), activation='relu', input_shape=(28, 28, 1)),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(128, activation='relu'),
    layers.Dense(10, activation='softmax')
])

# Using Adam optimizer and Sparse Categorical Cross Entropy [cite: 17]
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# Training for 5 epochs to reach the observed 97% accuracy [cite: 24, 27, 33]
print("Starting training...")
model.fit(x_train, y_train, epochs=5, validation_data=(x_test, y_test))

# 3. Conversion to TFLite
# Converting for edge deployment to minimize model footprint [cite: 18, 19]
converter = tf.lite.TFLiteConverter.from_keras_model(model)
tflite_model = converter.convert()
```

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# Saving as model.tflite [cite: 37]
with open('model.tflite', 'wb') as f:
    f.write(tflite_model)

# Final size check (Expected: ~350 KB) [cite: 38, 49]
size_kb = os.path.getsize('model.tflite') / 1024
print(f"Model successfully converted! Final Size: {size_kb:.2f} KB")

```

2026-02-13 00:17:00.495709: I external/local_xla/xla/tsl/cuda/cudart_stub.cc:31]
 Could not find cuda drivers on your machine, GPU will not be used.
 2026-02-13 00:17:00.564207: I tensorflow/core/platform/cpu_feature_guard.cc:210]
 This TensorFlow binary is optimized to use available CPU instructions in
 performance-critical operations.

To enable the following instructions: AVX2 AVX512F FMA, in other operations,
 rebuild TensorFlow with the appropriate compiler flags.

2026-02-13 00:17:04.409554: I external/local_xla/xla/tsl/cuda/cudart_stub.cc:31]
 Could not find cuda drivers on your machine, GPU will not be used.

Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz>

11490434/11490434 0s

0us/step

/home/a70a1744-edbd-4968-80f1-5208da2d0c32/.local/lib/python3.11/site-
 packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not
 pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential
 models, prefer using an `Input(shape)` object as the first layer in the model
 instead.

super().__init__(activity_regularizer=activity_regularizer, **kwargs)

2026-02-13 00:17:08.296698: E

external/local_xla/xla/stream_executor/cuda/cuda_platform.cc:51] failed call to
 cuInit: INTERNAL: CUDA error: Failed call to cuInit: UNKNOWN ERROR (303)

Starting training...

Epoch 1/5

2026-02-13 00:17:09.113775: W

external/local_xla/xla/tsl/framework/cpu_allocator_impl.cc:84] Allocation of
 188160000 exceeds 10% of free system memory.

1875/1875 22s 11ms/step -

accuracy: 0.9564 - loss: 0.1463 - val_accuracy: 0.9748 - val_loss: 0.0738

Epoch 2/5

1875/1875 19s 10ms/step -

accuracy: 0.9845 - loss: 0.0517 - val_accuracy: 0.9849 - val_loss: 0.0453

Epoch 3/5

1875/1875 24s 13ms/step -

accuracy: 0.9896 - loss: 0.0334 - val_accuracy: 0.9841 - val_loss: 0.0435

Epoch 4/5

```

1875/1875          19s 10ms/step -
accuracy: 0.9930 - loss: 0.0221 - val_accuracy: 0.9851 - val_loss: 0.0452
Epoch 5/5
1875/1875          19s 10ms/step -
accuracy: 0.9952 - loss: 0.0147 - val_accuracy: 0.9870 - val_loss: 0.0447
INFO:tensorflow:Assets written to: /tmp/tmplf0tnr_3/assets
INFO:tensorflow:Assets written to: /tmp/tmplf0tnr_3/assets
Saved artifact at '/tmp/tmplf0tnr_3'. The following endpoints are available:

* Endpoint 'serve'
  args_0 (POSITIONAL_ONLY): TensorSpec(shape=(None, 28, 28, 1),
dtype=tf.float32, name='keras_tensor')
Output Type:
  TensorSpec(shape=(None, 10), dtype=tf.float32, name=None)
Captures:
  129053765230544: TensorSpec(shape=(), dtype=tf.resource, name=None)
  129053765231696: TensorSpec(shape=(), dtype=tf.resource, name=None)
  129053765231312: TensorSpec(shape=(), dtype=tf.resource, name=None)
  129053765229008: TensorSpec(shape=(), dtype=tf.resource, name=None)
  129053765232272: TensorSpec(shape=(), dtype=tf.resource, name=None)
  129053765231888: TensorSpec(shape=(), dtype=tf.resource, name=None)

WARNING: All log messages before absl::InitializeLog() is called are written to
STDERR
W0000 00:00:1770941932.058431      370 tf_tfl_flatbuffer_helpers.cc:364] Ignored
output_format.
W0000 00:00:1770941932.058455      370 tf_tfl_flatbuffer_helpers.cc:367] Ignored
drop_control_dependency.

Model successfully converted! Final Size: 2714.17 KB

2026-02-13 00:18:52.059022: I tensorflow/cc/saved_model/reader.cc:83] Reading
SavedModel from: /tmp/tmplf0tnr_3
2026-02-13 00:18:52.059978: I tensorflow/cc/saved_model/reader.cc:52] Reading
meta graph with tags { serve }
2026-02-13 00:18:52.059991: I tensorflow/cc/saved_model/reader.cc:147] Reading
SavedModel debug info (if present) from: /tmp/tmplf0tnr_3
I0000 00:00:1770941932.063888      370 mlir_graph_optimization_pass.cc:437] MLIR
V1 optimization pass is not enabled
2026-02-13 00:18:52.064536: I tensorflow/cc/saved_model/loader.cc:236] Restoring
SavedModel bundle.
2026-02-13 00:18:52.095179: I tensorflow/cc/saved_model/loader.cc:220] Running
initialization op on SavedModel bundle at path: /tmp/tmplf0tnr_3
2026-02-13 00:18:52.103229: I tensorflow/cc/saved_model/loader.cc:471]
SavedModel load for tags { serve }; Status: success: OK. Took 44213
microseconds.
2026-02-13 00:18:52.170513: I
tensorflow/compiler/mlir/tensorflow/utils/dump_mlir_util.cc:269] disabling MLIR

```

crash reproducer, set env var ``MLIR_CRASH_REPRODUCER_DIRECTORY`` to enable.

[]: